Personalized Recommendation with Contextual Bandits

NLP + CNN + RL

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COURSE: 6INTELSY FINAL PROJECT

Problem Statement & Motivation

The Problem

Traditional collaborative filtering struggles with the **"cold-start"** problem in news recommendation because content becomes outdated almost immediately.

The Solution

An intelligent system extracting semantic meaning from titles using **NLP + CNN** and dynamically adapting to user clicks via **RL**.

MVP: An offline simulation framework combining a 1D Text-CNN and a LinUCB bandit agent.

Dataset & Ethical Considerations

MIND-small

Microsoft News Dataset. EDA revealed massive class imbalance (CTR ~4.8%).

Privacy

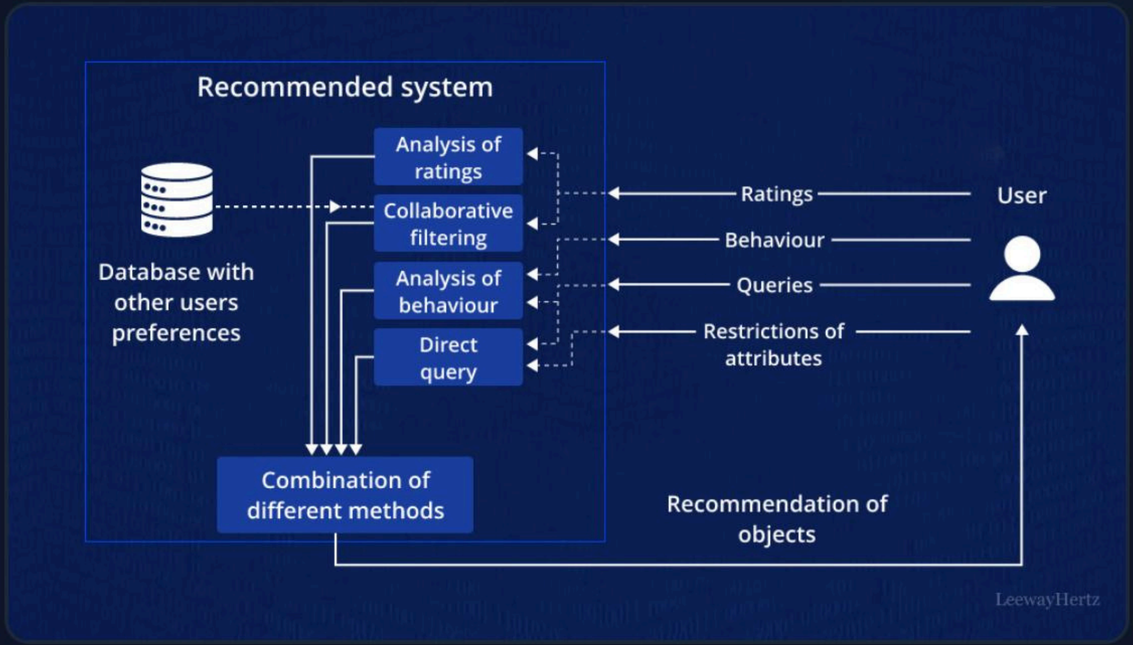
Strict adherence to anonymization. User IDs are hashed with no external data linkage.

Filter Bubbles

Exploration via Upper Confidence Bounds (UCB) prevents echo chambers.

System Architecture

- **NLP:** Pre-trained dense embeddings representing news vocabulary.
- **CNN:** 1D Text-CNN as a parallel feature extractor for local n-gram patterns.
- **RL:** Contextual Bandit (LinUCB) that utilizes CNN context for article selection.



NLP & CNN Components (Deep Learning)

1D Text-CNN Processing

Input consists of word embeddings of news titles.

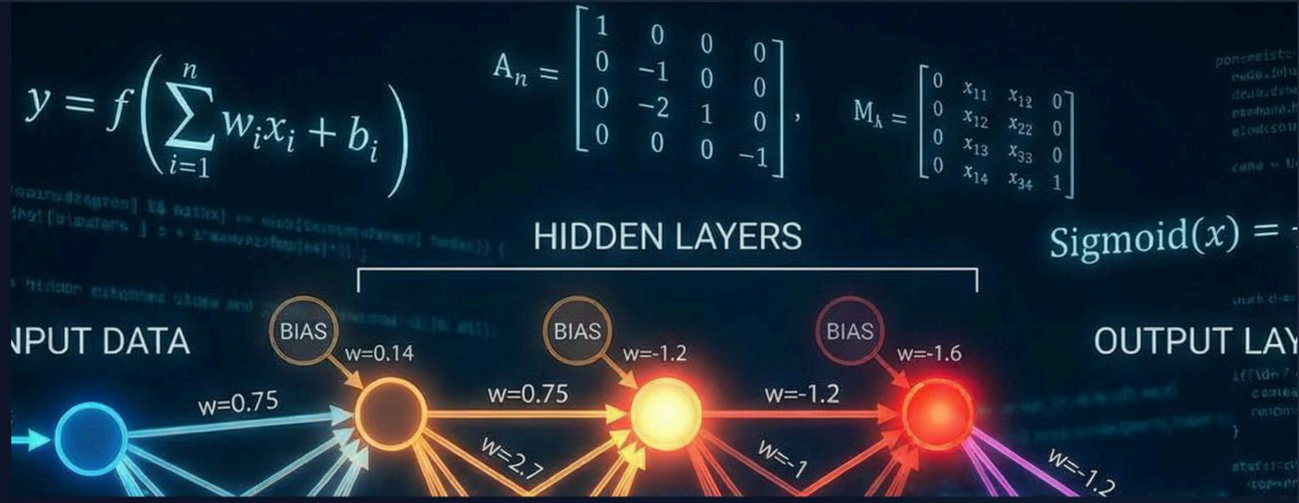
- **Filter Sizes:** 2, 3, and 4 to capture critical word relationships.
- **Global Max-Pooling:** Extracts the most salient features from each filter map.
- **Concatenation:** Results in a robust **128-dim** context vector.

Output Vector

128D

Captures core semantic meaning of the article.

Reinforcement Learning



Linear Contextual Bandits

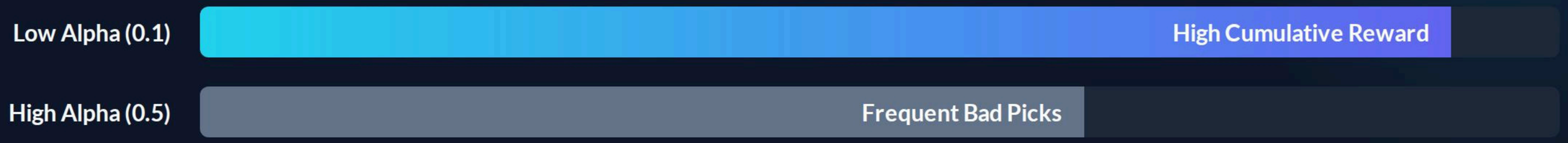
Agent: LinUCB Algorithm.

Mechanism: Observes context vector, calculates expected reward + uncertainty.

Environment: Offline Replay Simulator using MIND logs.

Reward: Binary (1 = Click, 0 = Skip).

Ablation 1: RL Exploration Rate α

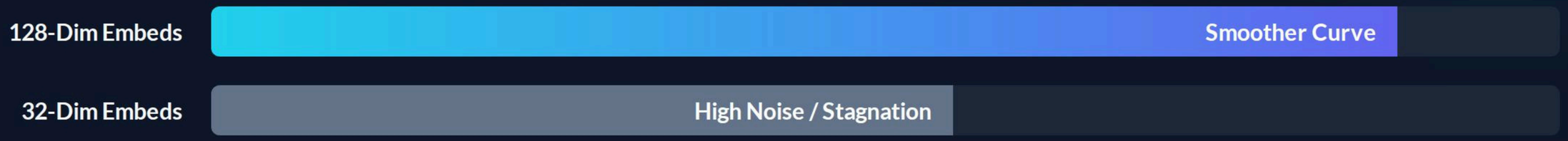


Finding: Optimal Alpha is 0.1

Because the environment is extremely sparse (4.8% CTR), excessive exploration forces the agent to pick low-quality articles too often, hindering convergence. **Faster convergence** was achieved with lower alpha.

Ablation 2: Embedding Dimensions

Convergence Trend Analysis (Terminal Reward Performance)



Conclusion: Richer semantic representations (128D) significantly aid the bandit in making accurate contextual matches between users and articles.

Error & Slice Analysis

News Category	CTR Performance	Reasoning
Finance	High	Specific, niche vocabulary with distinct semantic patterns.
Sports	Medium	Consistent entities, but highly dependent on seasonal interest.
Entertainment	Low	Subjective clickbait, semantic overlap, and unpredictable behavior.

Successfully Integrated Hybrid CNN-RL

Developed a robust recommender that extracts deep semantic features via 1D Text-CNN and balances exploration using LinUCB.

Thank You!

Questions & Discussion